

SIMATS ENGINEERING
ASSESSMENT TOOL 2
Case study

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions

Assessment Tool Weightage: 40%

Code of Conduct:

I C. K. Kandasamy (192421063) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

C. K. Kandasamy

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[Signature]

Analyze whether the problem is caused by the performance issue

Application layer:

* The performance issue is caused by both the transport protocol (UDP) and the application layer buffering strategy.

* Video conferencing application generally use UDP (User Datagram Protocol) because it provides fast transmission without waiting for acknowledgement.

* This is important for real time communication where delay is noticable and the packet loss

* Packet loss = 800

* Jitter = (80 ms)

* Fluctuating Bandwidth.

2. Justification

Real time multimedia

application require

↓ low latency

↓ low jitter

↓ continuous Streaming.

Recommended Improvements

Protocol - level

↓ Implement Quality of Service (QoS)

↓ Error Forward Error Correction (FEC)

↓ Enhance Adaptive bitrate Streaming.

Application - level

↓ Increase jitter buffer size

↓ Implement Adaptive buffering

↓ Automatically reduce video quality when bandwidth decreases.

Conclusion:

- ✓ The Conferencing problem is mainly due to poor CNP network condition, combined with insufficient buffering the application layer.
- ✓ Improving both networking quality and buffering mechanism will significantly improve user experience.

2) High DNS Resolution time

1. Reason for high DNS Resolution time

✓ the organization experience approximately 800ms DNS lookup time when delays webpage loading

• DNS server located far from user

• Heavy load on the DNS server.

α poor DNS caching

α Networking Congestion

↑ High latency between client and DNS Server

α repeated DNS lookup, due to improper

2. Proposed DNS Architecture

to reduce DNS lookup time below 50 ms, the organism should implement

Any cast DNS

α multiple DNS servers, share the same IP address
α hosts are automatically connected to nearest server.

DNS Prefetching

α frequently accessed +1 are resolved before any request them.

TTL Optimization

- Configure suitable TTL value so client cache

- DNS response longer.

- Store DNS record locally to reduce repeated queries.

3. Advantage and Limitation

Advantage:

- faster DNS resolution

- Improved scalability

Limitation:

- higher deployment cost

- more complex configuration.

Conclusion:

Using Anycast DNS, DNS caching, DNS prefetching and optimized TTL values can reduce DNS resolution time from 800 ms to below 50 ms.

3) Centralized DB Architecture:

Impact of centralized DB

In the current system, all DB requests are directed to single primary DB server.

This causes:

- high server workload
- increased backup latency
- single point of failure
- poor scalability during heavy traffic

2. Distributed DB Solution:

A distributed architecture should include:

- multiple DB server across different geographical regions

Any client having to direct user to nearest server.

3. Effect of TTL, Anycast, and DNI Caching

TTL (time to live)

• determines how long DNS records remain in

cache

• Reduce repeated DNS queries.

Anycast Routing

• Send users to the nearest DNS server

• Reduce latency and improve response time.

DNS Caching

• Store frequently requested records

• Reduce server workload.

• Improve response speed.

Conclusion:

A distributed DNS architecture combined with Anycast routing, DNS caching and proper TTL improves performance, scalability and lookup latency.

W)

Electronic Stock trading platform =

1. tcp-related factors causing latency

• The trading platform experiences an increase in acknowledgement latency time to time due to several tcp-related factors

a) Flow control

• tcp limits the amount of data sent based on available buffer

• During heavy trading, the receiver may become congested, slowing data in

transmission.

b) Nagle Algorithm

Nagle Algorithm combines small packets before transmission network traffic

• Although efficient, introduces additional delay when establishing connection for high frequency trading.

c) SYN Queue Limitation

When many users attempt to establish TCP connection simultaneously, the SYN queue becomes full, delaying or rejecting new connection requests.

2) Effect on high frequency trading

• delayed order confirmation

• slower execution of buy and sell orders

• increased transaction latency

• reduced customer satisfaction.

3. Recommended TCP configuration changes

• Implement Nagle's Algorithm using TCP_NODELAY

• Increase TCP send and receive buffer sizes

• Increase SYN queue capacity

• Optimize congestion control mechanisms

• Use TCP Fast Open to reduce connection establishment time

• Maintain persistent TCP connection to avoid repeated handshakes.

Conclusion:

Optimizing TCP settings, increasing buffer capacity and improving management can significantly reduce latency and improve the reliability and performance of electronic stock trading platform.